RailTel Safety Solutions: Enhancing Railway Safety Using IoT, AI, and Lidar Technologies

Simhadri Vishwathma
Computer Science Engineering(Artificial Intelligence & Machine Learning),
Keshav Memorial Engineering College,
Hyderabad,India.
vishwathma2@gmail.com

Mr. Pegada Naresh Kumar
Department of Computer Science Engineering (Artificial Intelligence & Machine Learning) ,
Keshav Memorial Engineering College,
Hyderabad,India
pnrshkumar@gmail.com

Abstract-Railway transportation in India remains the most affordable and widely used mode for both long-distance and suburban travel. However, safety issues such as undetected cracks in railway tracks and unexpected obstacles remain major causes of accidents. To address this, we propose an integrated, dynamic, and autonomous railway monitoring system that combines crack detection and obstacle avoidance using modern sensor technology. The system leverages GPS modules to track geographical coordinates and send real-time alert messages in case of any detected threats. It employs Arduino-based microcontrollers, such as ATmega2560, to control and coordinate various components including Sharp IR sensors for obstacle detection and motor drivers for path correction. This dual-functional solution aims to enhance safety by preventing derailments due to cracks and collisions due to unexpected obstacles, offering a robust and scalable model for smart railway infrastructure.

Keywords:
Railway Track Crack Detection; Obstacle Avoidance; GPS Module; Arduino Microcontroller; ATmega2560; Sharp IR Sensor; Motor Driver; Autonomous Vehicle; Real-time Alert System; Railway Safety; Embedded Systems; Smart Transportation; Railway Monitoring System

i. INTRODUCTION

Railways are one of the most affordable and widely used modes of transportation in India, especially for long-distance and suburban travel. However, safety remains a critical concern due to frequent accidents caused by cracks in railway tracks and unexpected obstacles. Traditional inspection methods are often manual and time-consuming, making them inefficient for large-scale monitoring. To address these challenges, there is a growing need for automated systems that can detect faults and enhance railway safety.

This project proposes an integrated solution that combines railway track crack detection with obstacle avoidance functionality. The system utilizes Arduino-based microcontrollers, GPS modules for location tracking, and sharp IR sensors for obstacle detection. By implementing real-time monitoring and autonomous navigation, the system not only identifies potential hazards but also communicates their exact location. This ensures timely alerts and preventive actions, contributing to a safer and more reliable railway transportation system

ii. Literature Summary

1. "Railway Track Crack Detection" (IJCRT, May 2023)
Proposed a basic system using sensors for detecting cracks in real-time. It faced issues with limited data processing and communication. The improvement involved ESP32 with IR and ultrasonic sensors for real-time alerting.

2. "Efficient Railway Track Crack Detection System" (IJRT, 2023)
Suggested a system for railway crack detection, but lacked instant alert mechanisms. Overcame this by integrating GSM and GPS modules for faster notifications to authorities.

3. "Intelligent Railway Track Crack and Obstacle Detection Using IoT" (IRJMETS, May 2023)
The system had limited centralized sensor control. The enhanced version used ESP32 and ultrasonic sensors with Telegram notifications for better alerting and control.

4. "Advance Railway Track Crack Detection System Using IoT" (IJNRD, May 2023)
Highlighted issues with scalability across large railway networks. Improved with a scalable IoT-based design using IR and GSM/GPS modules for detection and reporting.

5. "Railway Track Crack & Object Detection Using GSM & GPS" (IJRES, 2022)
Faced challenges in obstacle detection due to poor sensor accuracy. Solved by using IR sensors and GSM/GPS for concurrent obstacle and crack detection.

6. "Automatic Railway Track Crack Detection System Using Ultrasonic Sensors" (IRJET, March 2023)
Missed real-time location tracking of cracks. Overcome by including GSM for live crack detection data transmission with ultrasonic sensors.

7. "Railway Track Crack and Obstacle Detector" (ResearchGate, 2019)
Lacked monitoring and reporting capabilities. Upgraded with ultrasonic ranging for better obstacle detection and track monitoring.

8. "Broken Rail Detection With Texture Image Processing Using GLCM" (arXiv, April 2023)
Relied heavily on image quality, which limited effectiveness. Solution was the GLCM method for texture-based crack identification.

9. "Railway Track Crack Detection & Obstacle Detection System" (IRJMETS, March 2022)
Identified limited obstacle detection and reporting. Used IR sensors and GPS-enabled Arduino Uno to improve accuracy and localization.

10. "Development of Railway Track Crack Detection System Using Arduino" (ResearchGate, 2023)
Suffered from poor sensor integration. Enhanced with Arduino, GPS, and ultrasonic sensors for comprehensive crack tracking.

11. "Automated Crack Detection of Railway Track" (IOSR-JEEE, 2020)
Prone to false positives during detection. Solved using Bayesian analysis to refine signal detection via ultrasonic and PIR sensors.

TABLE 1

Study	Presented Work	Limitation	Overcome
Railway Track Crack Detection (IJCRT, May 2023)	Integrated IR and ultrasonic sensors with ESP32	Limited real-time communication and data processing	Enabled real-time crack detection and alert systems
Efficient Railway Track Crack Detection System (IJRT, 2023)	System with GSM and GPS modules	Lack of robust alert mechanism	Instant alerts to authorities
Intelligent Railway Track Crack and Obstacle Detection Using IoT (IRJMETS, May 2023)	Real-time detection with Telegram alerts	Limited sensor integration with centralized control	Centralized control and alerts
Advanced Railway Track Crack Detection Using IoT (IJNRD, May 2023)	IoT-based scalable crack detection	Limited scalability	Scalability with IR, GSM/GPS modules
Railway Track Crack & Object Detection Using GSM & GPS (IJRES, 2022)	Robot integrating sensors for detection	Ineffective obstacle detection	Improved simultaneous crack and obstacle detection
Automatic Railway Track Crack Detection Using Ultrasonic Sensors (IRJET, March 2023)	GSM-enabled crack tracking	No real-time location tracking	Real-time data transmission

Railway Track Crack and Obstacle Detector (ResearchGate, 2019)	Ultrasonic for obstacle & track continuity	Limited reporting & monitoring	Enhanced reporting
Broken Rail Detection Using GLCM (arXiv, 2023)	Texture image processing for cracks	Image quality dependency	Texture-based defect detection
Railway Track Crack Detection & Obstacle System (IRJMETS, 2022)	IR + Arduino + GPS detection	Poor obstacle identification	Accurate location detection
Development of Railway Track Crack Detection Using Arduino (ResearchGate, 2023)	Sensor-integrated Arduino model	Limited sensor control	Better integration and efficiency
Automated Crack Detection of Railway Track (IOSR-JEEE, 2020)	Bayesian filter for false positives	High false positives	Improved signal filtering

Early models struggled with issues like limited data processing, poor real-time alerts, sensor inaccuracy, and scalability challenges. Over time, improvements included the integration of **ESP32, GSM/GPS modules, ultrasonic and IR sensors**, and **Telegram/GSM notifications** for real-time monitoring and alerting. Advanced approaches also incorporated **image processing (GLCM)** and **Bayesian analysis** to enhance crack detection accuracy and reduce false positives. Overall, the trend shows progressive enhancement in detection precision, alert mechanisms, and system scalability for safer rail operations.

TABLE 2

Study	Real-time Detection	Sensor Fusion	Obstacle Detection	Location Tracking	Alert System	IoT Enabled	Cloud Storage	Efficiency	Scalability	Cost-effective
IJCRT (2023)	✓	✓	-	-	✓	✓	-	✓	-	✓
IJRT (2023)	✓	✓	-	✓	✓	✓	-	✓	-	✓
IRJMETS (May 2023)	✓	✓	✓	-	✓	✓	-	✓	✓	✓
IJNRD (2023)	✓	✓	-	✓	✓	✓	-	✓	✓	✓
IJRES (2022)	✓	✓	✓	✓	✓	✓	-	✓	✓	✓
IRJET (2023)	✓	✓	-	-	✓	✓	-	✓	-	✓
ResearchGate (2019)	✓	✓	✓	-	✓	-	-	✓	-	✓
arXiv (2023)	✓	-	-	-	-	-	✓	✓	-	-
IRJMETS (2022)	✓	✓	✓	✓	✓	✓	-	✓	-	✓
ResearchGate (2023)	✓	✓	✓	✓	✓	✓	-	✓	✓	✓
IOSR-JEEE (2020)	✓	✓	-	-	✓	-	-	✓	-	-

iii. Related Work

In recent years, several researchers have focused on developing sensor-based systems to improve safety in autonomous vehicles and railway infrastructure. One of the key challenges in autonomous navigation is obstacle detection and avoidance. Studies such as those by Adarsh et al. (2016) have explored the comparative performance of ultrasonic and infrared (IR) sensors across various surface materials. Their results showed that while both sensors are useful, ultrasonic sensors provide better reliability over longer distances. Other researchers, such as Sachin Modi (2002), emphasized the importance of sensor arrangement, including fixed sonar sensors and rotating systems, in enhancing obstacle detection accuracy. These insights have contributed to the growing adoption of multi-sensor systems for robust autonomous vehicle design.

The use of microcontrollers, especially Arduino boards like ATmega2560 and Uno, has been instrumental in these systems. In the obstacle detection project, IR sensors are integrated with Arduino and motor drivers to guide vehicles autonomously when an obstacle is encountered. Advanced simulations using MATLAB and Simulink have enabled engineers to develop and test control systems efficiently before physical deployment. Simulation models such as those proposed by Jie et al. (2009) establish a closed-loop control between sensors and actuators to respond to environmental changes dynamically, further increasing the feasibility of automated obstacle avoidance in practical scenarios.

Parallel to developments in obstacle detection, researchers have also addressed the problem of crack detection in railway tracks. Traditional manual inspection methods have been largely replaced by automated mobile units equipped with ultrasonic sensors and GPS modules. The crack detection system described by Arun Kumar et al. (2020) demonstrates a real-time approach where any detected fault is relayed immediately to a nearby control station using GSM or SMS alerts. This approach overcomes earlier challenges seen in LED-LDR-based systems, which required precise alignment and were highly affected by lighting and environmental conditions. These newer systems also incorporate motor drivers to halt the vehicle in case of a detected crack, thereby adding a safety mechanism.

Moreover, the literature shows a trend toward combining crack detection with obstacle recognition on railway tracks to create multifunctional monitoring robots. The integration of IR and ultrasonic sensors allows these systems to differentiate between cracks and physical obstructions on tracks. GPS modules play a vital role by enabling accurate localization, while microcontrollers manage signal processing and control logic. The inclusion of wireless technologies such as GSM, and emerging protocols like Wireless Application Protocol (WAP), has enhanced communication reliability. These innovations collectively highlight the shift from basic automation toward intelligent, scalable, and cost-effective systems that ensure both passenger safety and infrastructure resilience.

iv. Proposed Work

The proposed system aims to develop an integrated solution that combines both obstacle detection and railway track crack detection into a single autonomous unit. The core components of the system include an Arduino-based microcontroller, ultrasonic and infrared (IR) sensors, GPS module, and a motor driver circuit. The ultrasonic sensor is primarily responsible for identifying cracks or gaps in the railway tracks, while the IR sensors are deployed for detecting physical obstacles in the path. The system is mounted on a moving robotic platform that traverses the railway track, continuously monitoring for both surface anomalies and unexpected obstructions.

Upon detecting a crack or obstacle, the sensor data is processed by the microcontroller, which simultaneously triggers two responses: it halts the vehicle to avoid further damage or accidents, and it sends real-time geographic coordinates to the nearest control station using a GPS module. This ensures that corrective action can be taken promptly by railway authorities. The motor driver enables controlled movement of the vehicle, including directional changes, which can be adjusted dynamically based on the type of obstacle detected. The servo motor is used to rotate sensors for scanning a wider area in front of the vehicle, improving detection range and accuracy.

To enhance the reliability and efficiency of the system, the proposed design also includes the potential integration of cloud-based data logging for historical analysis and predictive maintenance. Sensor readings and location data can be stored and analyzed to detect recurring fault patterns or risky areas on the track. This multi-sensor, microcontroller-driven system offers a scalable, cost-effective solution that enhances the safety and automation of railway operations. Future improvements could include sensor fusion with LIDAR or AI-based image recognition to further boost detection accuracy and adaptability under varying environmental conditions.

v. Result and Implementation

The implementation of the proposed system was carried out using an Arduino Mega 2560 microcontroller as the central control unit, interfaced with IR sensors for obstacle detection and ultrasonic sensors for crack detection. A GPS module was integrated to obtain real-time location data, which was transmitted via serial communication to simulate alert delivery to control centers. The system was mounted on a wheeled robotic platform, powered by a dual-shaft DC geared motor and controlled through an L298N motor driver. Servo motors were employed to rotate sensors for wider scanning angles. The entire system was tested on a model railway track to evaluate sensor responsiveness, signal processing, and actuation efficiency.

The results demonstrated accurate detection of both static obstacles and track discontinuities. When a crack was identified by the ultrasonic sensor or an obstacle was detected by the IR sensor, the microcontroller immediately halted the vehicle and retrieved the GPS coordinates of the fault location. This validated the system's capability for real-time fault identification and location tracking. The prototype functioned reliably in simulated environments and proved to be energy-efficient and responsive. These outcomes indicate that the proposed system can be effectively deployed for railway safety monitoring, with the potential for future upgrades like cloud connectivity or AI-based detection modules.

References

[1] A. Rizvi, P. Khan and D. Ahmad, "Crack Detection In Railway Track Using Image Processing", International Journal of Advance Research, Ideas and Innovations in Technology., vol. 3, no. 4, 2017.

[2] S. Srivastava, R. Chaurasia, S. Abbas, P. Sharma and N. Singh, "Railway Track Crack Detection Vehicle", International Advanced Research Journal in Science, Engineering and Technology, vol. 4, no. 2, pp. 145-148, 2017.

[3] K.Bhargavi and M. Janardhana Raju "Railway Track Crack Detection Using Led-Ldr Assembly", International Journal of Advanced Research in Electronics and Communication Engineering (IJARECE), vol. 3, no. 9, pp. 1230-1234, 2014.

[4] B. Siva Ram Krishna, D. Seshendia, G. Govinda Raja, T. Sudharshan and K. Srikanth, "Railway Track Fault DetectionSystem By Using IR Sensors And Blutooth Technology", Asian Journal of Applied Science and Technology (AJAST), vol. 1, no. 6, pp. 82-84, 2017.

[5] [5] P.Navaraj, "Crack Detection System For Railway Track By Using Ultrasonic And Pir Sensor", vol. 1, no. 1, pp. 126-130, 2014. [6] [6] D.Narendhar Singh and D. Naresh, "Railway Track Crack Detection And Data Analysis", vol. 5, no. 4, pp. 1859-1863, 2017.

[7]2017. Available: https://m.timesofindia.com/india/586 train-accidents-in-last-5-years-53-due-to derailments/amp_articleshow/60141578.cms.

[8] Kaleemuddin S , and Bose K. (2016). Performance comparison of Infrared and Ultrasonic sensors for obstacle of different materials in vehicles/robot navigation applications, IOP Conf. Series: Materials Science and Engineering, 10.1088/1757-899X/149/1/012141.

[9] Modi.(2002). Comparison of three obstacle avoidance methods for an autonomous guided vehicle. Used

[10] G, Patange, S, A, & J.(2017). Characteristics of different sensors for Distance,Measurement.International Research Journal of Engineering and Technology (IRJET) 698-701.

[11] Varghese & Boone. (2015). Overview of Autonomous Vehicle Sensors and Systems, Proceedings of the 2015 International Conference on Operations Excellence and Service Engineering 178.

[12] Domínguez, E, J, J, & C.(2011). LIDAR based Perception Solution for Autonomous Vehicles2011,11th International Conference on Intelligent Systems Design and Applications 790.

[13] Zhou, Chen, & Xiu.(2009). A Simulation Model to Evaluate and Verify Functions of autonomous vehicle based on simulink, Springer-Verlag Berlin Heidelberg, 10.1007/978 3-642-10817-4_64.

[14] Fernández, Domínguez, Fernández-Llorca, Alonso, & sotelo.(2013). Autonomous Navigation and obstacle avoidance of a Micro-bus, International Journal of Advanced Robotic Systems, 10.5772/56125.